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[28] 0s

```
model.fit(X_train, y_train)

prediction = model.predict(X_test)

mae = mean_absolute_error(y_test, prediction)
rmse = root_mean_squared_error(y_test, prediction)

print("===== MODEL EVALUATION =====\n")
print("Mean Absolute Error :", round(mae, 2))
print("Root Mean Squared Error :", round(rmse, 2))

print("\nTask 5.4 Completed Successfully.")
```

```
... ===== MODEL EVALUATION =====

Mean Absolute Error : 152.21
Root Mean Squared Error : 237.39

Task 5.4 Completed Successfully.
```

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```
[28] ✓ Os
# LEVEL 5 - TASK 5.4
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_absolute_error, root_mean_squared_error
import pandas as pd

df = pd.read_csv("Modeling_Dataset.csv")

X = df[["Total_Distance", "Number_of_Stops"]]
y = df["Journey_Duration"]

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

model = LinearRegression()

model.fit(X_train, y_train)

prediction = model.predict(X_test)

mae = mean_absolute_error(y_test, prediction)
rmse = root_mean_squared_error(y_test, prediction)

print("===== MODEL EVALUATION =====\n")
print("Mean Absolute Error :", round(mae, 2))
print("Root Mean Squared Error :", round(rmse, 2))

print("\nTask 5.4 Completed Successfully.")
```

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[27] 0s

```
"Predicted": predictions
})

print(result.head(10))

result.to_csv("Prediction_Result.csv", index=False)

print("\nLinear Regression Model Built Successfully.")
print("\nTask 5.3 Completed Successfully.")

... ===== SAMPLE PREDICTIONS =====

   Actual   Predicted
0   690.0   368.707129
1   180.0   262.128350
2    55.0   195.543279
3   190.0   129.658595
4   113.0   246.748817
5    38.0    94.992887
6   450.0   292.489264
7    59.0    92.500590
8   705.0   707.352075
9  1185.0   526.251424

Linear Regression Model Built Successfully.

Task 5.3 Completed Successfully.
```

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CommandsCodeTextRun all

[27] ✓ 0s

```
# LEVEL 5 - TASK 5.3
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
import pandas as pd

df = pd.read_csv("Modeling_Dataset.csv")

x = df[["Total_Distance", "Number_of_Stops"]]
y = df["Journey_Duration"]

X_train, X_test, y_train, y_test = train_test_split(
    x,
    y,
    test_size=0.2,
    random_state=42
)

model = LinearRegression()

model.fit(X_train, y_train)

predictions = model.predict(X_test)

print("***** SAMPLE PREDICTIONS *****\n")

result = pd.DataFrame({
    "Actual": y_test.values,
    "Predicted": predictions
})

print(result.head(10))
```

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Modeling dataset created successfully.

Task 5.1 Completed Successfully.

[26] 1s

```
# LEVEL 5 - TASK 5.2
from sklearn.model_selection import train_test_split
import pandas as pd

df = pd.read_csv("Modeling_Dataset.csv")

X = df[["Total_Distance", "Number_of_Stops"]]
y = df["Journey_Duration"]

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

print("Training Data Shape :", X_train.shape)
print("Testing Data Shape :", X_test.shape)

print("\nTask 5.2 Completed Successfully.")
```

Training Data Shape : (8890, 2)
Testing Data Shape : (2223, 2)

Task 5.2 Completed Successfully.

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===== MODELING DATASET =====

```
***
  Train_No  Total_Distance  Number_of_Stops  Journey_Duration
0         107             78                4             105.0
1         108             83                4             115.0
2         128            978               22            1325.0
3         290           2694               14             480.0
4         401           1618               12             750.0

===== DATASET INFORMATION =====

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11113 entries, 0 to 11112
Data columns (total 4 columns):
#   Column                Non-Null Count  Dtype
---  ---                ---
0   Train_No              11113 non-null  int64
1   Total_Distance        11113 non-null  int64
2   Number_of_Stops       11113 non-null  int64
3   Journey_Duration      11113 non-null  float64
dtypes: float64(1), int64(3)
memory usage: 347.4 KB
None

Modeling dataset created successfully.

Task 5.1 Completed Successfully.
```

===== DATASET INFORMATION =====

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```
[25] ✓ 0s # LEVEL 5 - TASK 5.1
import pandas as pd

features = pd.read_csv("Feature_Engineered_Dataset.csv")
journey = pd.read_csv("Journey_Duration.csv")

journey["Journey_Duration"] = pd.to_timedelta(
    journey["Journey_Duration"],
    errors="coerce"
).dt.total_seconds() / 60

model_data = pd.merge(
    features,
    journey[["Train_No", "Journey_Duration"]],
    on="Train_No"
)

model_data = model_data.dropna()

print("===== MODELING DATASET =====\n")
print(model_data.head())

print("\n===== DATASET INFORMATION =====\n")
print(model_data.info())

model_data.to_csv("Modeling_Dataset.csv", index=False)

print("\nModeling dataset created successfully.")
print("\nTask 5.1 Completed Successfully.")

... ===== MODELING DATASET =====
```

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